

m training examples, n features.

Gradient Descent

- * choose α
- * need iteration

$n \sim 10^6$

Normal Eq.

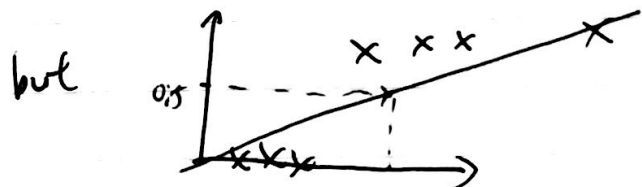
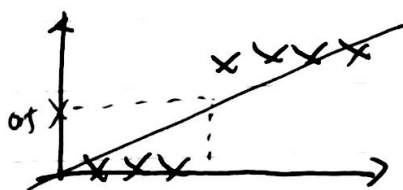
- * no need to choose α
- * no need to iterate
- * compute $(X^T X)^{-1}$ ($n \times n$ or $(n+1) \times (n+1)$)
slow! $O(n^3)$
 $n \sim 10000$, still ok
if consider θ_0

* What if $(X^T X)^{-1}$ is non-invertible?

- ① Redundant features (linearly dependent)
 $x_1, x_2, x = (3.28)^2 x_2$, delete x_1 or x_2
- ② Too many features ($m \leq n$)
→ Delete some features, or else regularization.

Logistic Regression $y=0, y=1$ → classification

example why not using linear Regression.



$y = 1$ if $\hat{h}(x) > 0.5$, $y = 0$ if $\hat{h}(x) < 0.5$
 $\hat{h}(x)$, also $\hat{h}(x)$ can be > 1 !
 < 0

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

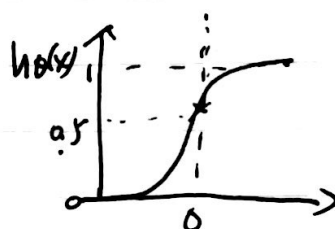
$$\Rightarrow h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

$h_{\theta}(x)$: estimated prob that $y=1$ on input x

eg. $x = \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} 1 \\ \text{tumor size} \end{pmatrix}$, $h_{\theta}(x) = 0.7$

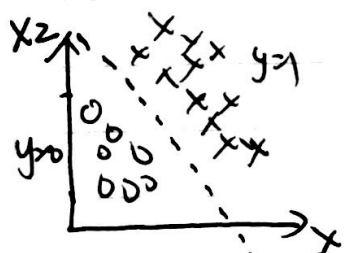
$\Rightarrow 70\%$ chance of tumor being malignant

$$\rightarrow h_{\theta}(x) = p(y=1 | x; \theta)$$



$$x > 0, h_{\theta}(x) > 0.5 \quad x < 0, h_{\theta}(x) < 0.5$$

Decision Boundary

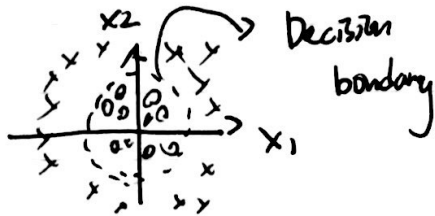


$$h_{\theta}(x) = g(\underbrace{\theta_0}_{-3} + \underbrace{\theta_1}_{+1}x_1 + \underbrace{\theta_2}_{+1}x_2)$$

predict "y=1" if $-3 + x_1 + x_2 \geq 0$

$$\rightarrow x_1 + x_2 \geq 3$$

non-linear decision boundary



$$h_{\theta}(x) = g \left(\overset{1}{\theta_0} + \overset{0}{\theta_1} x_1 + \overset{0}{\theta_2} x_2 + \overset{0}{\theta_3} x_1^2 + \overset{0}{\theta_4} x_2^2 \right)$$

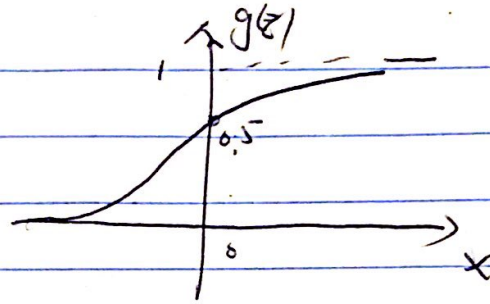
predict $\hat{y}=1$ if $x_1^2 + x_2^2 \geq 1$

Decision boundary is determined by hypothesis.

* logistic regression

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$



for classification

$$x > 0, g(z) > 0.5, y = 1$$

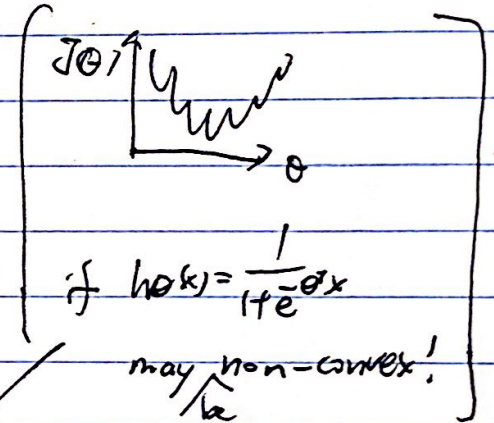
$$x < 0, g(z) < 0.5, y = 0$$

* linear regression :

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

* Cost function

$$\text{Cost}(h_{\theta}(x), y) = \frac{1}{2} (h_{\theta}(x) - y)^2$$



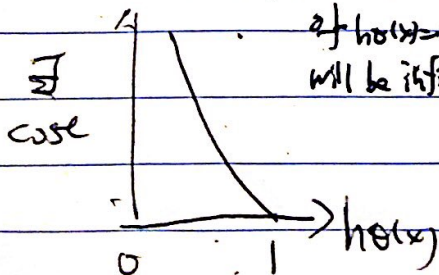
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)})$$

logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y=1, \text{ with } h_{\theta}(x)=1 \\ -\log(1-h_{\theta}(x)) & \text{if } y=0, \text{ with } h_{\theta}(x)=0 \end{cases}$$

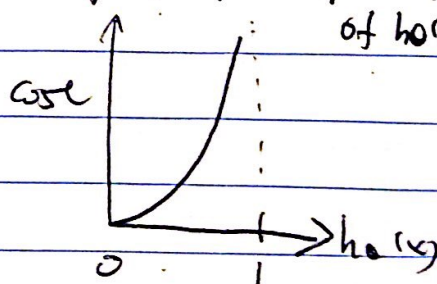
* if $y=1$, then penalty

if $h_{\theta}(x)=0$ will be infinite



* if $y=0$, then penalty

of $h_{\theta}(x)=1$ will be infinite



→ Picture why??

Cost($J(w)$) for logistic is $-(y \log(h_0(x)) + (1-y) \log(1-h_0(x)))$

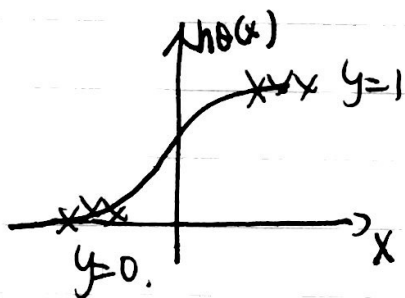
for linear Regression

Cost $\approx (h_0(x) - y)^2$ to minimize cost,

min cost $\Rightarrow h_0 = y$, get our hypothesis

for logistic Regression

Cost = $f(h_0(x))$.



(i) for $y=1$, cost is minimum, so $f(h_0(x)) = 0$
($=0$) but $h_0(x) \approx 1$

so $\log(h_0(x))$ a good guess
 $\approx \log(1) = 0$

(ii) for $y=0$, cost is minimum

so $f_2(h_0(x)) = 0$, $f_2(h_0(x) \approx 0) = 0$

$\log(1-h_0(x))$ is a good guess

but $h_0(x) < 1$, so $-\log(h_0(x))$ and $-\log(1-h_0(x))$

Check: for $y=1$, if $\hat{y}=0$, $\log(h_0(x))$ goes to $-\infty$ } can't minimize!
for $y=0$, if $\hat{y}=1$, $\log(1-h_0(x))$ goes to $-\infty$ } ~~can't~~ good

$$\text{Cost}(h_\theta(x), y) = -y \log(h_\theta(x)) - (1-y) \log(1-h_\theta(x))$$

$$h_\theta(x) = \frac{1}{1+e^{-x}}$$

$$= -y \log\left(\frac{1}{1+e^{-x}}\right) - (1-y) \cdot \log\left(\frac{e^{-x}}{1+e^{-x}}\right)$$

$$= -y \log\left(\frac{1}{1+e^{-x}}\right) - (1-y) \log\left(\frac{1}{e^x+1}\right)$$

$$= +y \log(1+e^{-x}) + (1-y) \log(1+e^x)$$

$$J(\theta) = \frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log(h_\theta(x^{(i)})) + (1-y^{(i)}) \log(1-h_\theta(x^{(i)})) \right]$$

$$\theta_j := \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j}$$

$$\frac{\partial (y \log(h_\theta(x)) + (1-y) \log(1-h_\theta(x)))}{\partial \theta_j}$$

$$= y \frac{\partial \log(h_\theta(x))}{\partial \theta_j} + (1-y) \frac{\partial \log(1-h_\theta(x))}{\partial \theta_j}$$

$$= y \cdot \frac{1}{h_\theta(x)} \cdot \frac{\partial h_\theta(x)}{\partial \theta_j} + (1-y) \cdot \left(\frac{1}{1-h_\theta(x)}\right) \cdot (-1) \frac{\partial h_\theta(x)}{\partial \theta_j}$$

$$= \left\{ \frac{y}{h_0(x)} - \frac{(1-y)}{1-h_0(x)} \right\} \cdot \frac{\partial h_0(x)}{\partial \theta_j}$$

$$h_0(x) = \frac{1}{1+e^{-\theta^T x}}, \quad \frac{\partial h_0(x)}{\partial \theta_j} = \frac{-1}{(1+e^{-\theta^T x})^2} \cdot e^{-\theta^T x} \cdot (-x_j);$$

$$= \frac{e^{-\theta^T x}}{(1+e^{-\theta^T x})^2} \cdot x_j$$

$$\left[\frac{1-h_0(x)}{e^{-\theta^T x}} = \frac{1}{e^{\theta^T x} + 1} \right]$$

$$= \left\{ y \cdot (1+e^{-\theta^T x}) - (1-y) \cdot (1+e^{\theta^T x}) \right\} \cdot \frac{\partial h_0(x)}{\partial \theta_j}$$

$$= \left\{ y \cdot (1+e^{-\theta^T x}) - (1-y) \cdot (1+e^{\theta^T x}) \right\} \cdot \frac{e^{-\theta^T x}}{(1+e^{-\theta^T x})^2} \cdot x_j$$

$$= \left(\frac{y e^{-\theta^T x}}{1+e^{-\theta^T x}} - \frac{(1-y) \cdot (e^{\theta^T x} + 1)}{(1+e^{-\theta^T x})^2} \right) \cdot x_j$$

$$= \left(\frac{y e^{-\theta^T x}}{1+e^{-\theta^T x}} - \frac{(1-y)}{(1+e^{-\theta^T x})} \right) \cdot x_j$$

$$\begin{cases} \text{linear: } h_0(x) = x^T \theta \\ \text{logit: } h_0(x) = \frac{1}{1+e^{-\theta^T x}} \end{cases}$$

$$= \left(\frac{y(1+e^{-\theta^T x}) - 1}{(1+e^{-\theta^T x})} \right) x_j = \left(y - \frac{1}{1+e^{-\theta^T x}} \right) \cdot x_j = (y - h_0(x)) x_j$$

$$\text{So } \frac{\partial J(\theta)}{\partial \theta_j} = (-1) \frac{1}{n} \cdot \sum_{i=1}^n (y^{(i)} - h_0(x^{(i)})) x_j^{(i)} = \frac{1}{n} \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

for logit cost fun, same as linear regression!!

* optimization algorithm

cost funn. $J(\theta)$. min $J(\theta)$
 θ

$\theta_j := \theta_j - \alpha \frac{\partial J}{\partial \theta_j} \rightarrow$ gradient descent

- conjugate gradient
- BFGs
- L-BFGs

Adv: No need to pick α manually
 faster than gradient descent.

Disad: More complex

"funn" for Qlave optimization algorithm

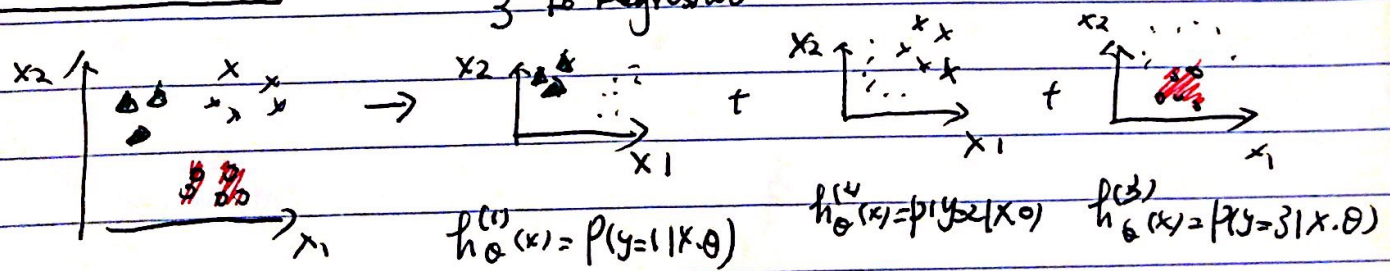
logistic Regression

$0 \leq h_{\theta}(x) < 1$. estimated probability to have $y=1$ with x, θ

$h_{\theta}(x) = P(y=1 | x, \theta)$

one vs. all

3 to Regression



One-vs-all

* Train i logistic Regression classifier $h_{\theta}^{(i)}(x)$ for each class i to predict the probability that $y=i$.

$$h_{\theta}^{(i)}(x) = p(y=i | x; \theta) \quad , \quad i=1, 2, 3, \dots$$

* On a new input x , to make a prediction, pick the class i that maximizes (check all classifiers)

$$\max_i h_{\theta}^{(i)}(x) \quad , \quad i=1, 2, 3, \dots$$

Logistic fun

8-5

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}}, \quad \text{to fit } \vec{\theta} = (\theta_1, \theta_2, \dots)$$

using maximum likelihood

$$= \frac{1}{1 + \frac{1}{e^{\theta^T x}}}$$

$$= \frac{1}{1 + \frac{1}{\text{odds}}} \rightarrow \text{odds} = e^{\theta^T x}$$

$$\text{cost} = - \frac{1}{n} \sum_{i=1}^n y_i \log(h_{\theta}(x^{(i)})) + (1 - y_i) \log(1 - h_{\theta}(x^{(i)}))$$

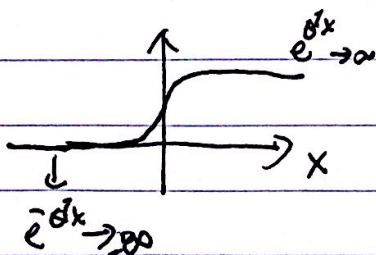
if $y^{(i)} = 1$, to minimize cost, $\log(h_{\theta}(x^{(i)})) \rightarrow 0$

$$\Rightarrow h_{\theta}(x^{(i)}) = 1 = \frac{1}{1 + e^{-\theta^T x}} \Rightarrow e^{-\theta^T x} \rightarrow 0 \Rightarrow e^{\theta^T x} \rightarrow \infty$$

$\frac{1}{e^{\theta^T x}} \rightarrow 0$ as x large

if $y^{(i)} = 0$, to minimize cost $\log(1 - h_{\theta}(x^{(i)})) \rightarrow 0$

$$\Rightarrow h_{\theta}(x^{(i)}) = 0 = \frac{1}{1 + e^{-\theta^T x}} \Rightarrow e^{-\theta^T x} \rightarrow \infty \rightarrow -x \text{ large}$$



$$\left[\begin{array}{l} f(x) = \beta_0 + \beta_1 x \quad \text{Linear Regression} \rightarrow \beta_1 \text{ gives average change} \\ \text{Odds} = e^{\theta^T x} \rightarrow \log(\text{odds}) = \beta_0 + \beta_1 x \quad \text{Logistic Regression} \end{array} \right. \begin{array}{l} \text{with one-unit increase} \\ \text{in } x \end{array}$$

$\rightarrow \beta_1$: average change of $\log(\text{odds})$
with one-unit change in x

* likelihood fun:

$$l(\beta_0, \beta_1) = \prod_{i: y_i=1} p(x_i) \cdot \prod_{i: y_i=0} (1 - p(x_i))$$

$$\frac{p(x)}{1 - p(x)} = e^{\theta^T x} \rightarrow p(x) = e^{\theta^T x} - e^{\theta^T x} p(x)$$

$$\rightarrow (1 + e^{\theta^T x}) p(x) = e^{\theta^T x}$$

$$\rightarrow p(x) = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}} = \frac{1}{1 + e^{-\theta^T x}} = h(\theta^T x)$$

* In Statistical learning Book,
logistic Reg is to maximize likelihood.

* In Andrew's course, we minimize Cost fun

$$\text{So } h_{\theta}(x) = p(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots)}}$$

hypothesis or probability of observing x on $y=1$.

$|x| = \text{hodd}$

$$l(\theta_0, \theta_1, \dots, \theta_n) = \prod_{i: y_i=1} P(x_i) \cdot \prod_{j: y_j=0} (1 - P(x_j))$$

$$\rightarrow \log l = \sum_{i: y_i=1} \log P(x_i) + \sum_{j: y_j=0} \log (1 - P(x_j))$$

$$\rightarrow -\log l = C = - \sum_i y_i \log P(x_i) - \sum_i (1 - y_i) \log (1 - P(x_i))$$

$$\rightarrow \text{maximize } l \rightarrow \text{minimize } -\log C \quad \checkmark$$

$$\text{note } \log \left(\frac{p(x)}{1-p(x)} \right) = \log \text{ odds} = \log \left(\frac{1}{e^{-\theta^T x}} \right) = \theta^T x$$

* Binary Classification ($y=0, 1$)

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = p(y=1 | \theta, x) = \frac{e^{\theta^T x}}{1 + e^{\theta^T x}}$$

$$C(\theta) = - \sum_{i=1}^m \left(y_i \log h_{\theta}(x_i) + (1 - y_i) \log (1 - h_{\theta}(x_i)) \right)$$

* Multi-class classification ($y=1, 2, \dots, k$) (Softmax Regression)

$$h_{\theta}(x) = \begin{bmatrix} p(y=1 | x, \theta) \\ p(y=2 | x, \theta) \\ \vdots \\ p(y=k | x, \theta) \end{bmatrix} = \frac{1}{\sum_{j=1}^k e^{\theta^{(j)T} x}} \begin{bmatrix} e^{\theta^{(1)T} x} \\ e^{\theta^{(2)T} x} \\ \vdots \\ e^{\theta^{(k)T} x} \end{bmatrix}$$

↑
normalized

$$\vec{\theta}^{(1)}, \vec{\theta}^{(2)}, \dots, \vec{\theta}^{(k)} \in \mathbb{R}^n \text{ (n predictors)}$$

$$\Theta = \begin{bmatrix} \begin{bmatrix} 1 \\ \theta^{(1)} \end{bmatrix} & \begin{bmatrix} 1 \\ \theta^{(2)} \end{bmatrix} & \dots & \begin{bmatrix} 1 \\ \theta^{(k)} \end{bmatrix} \end{bmatrix}$$

i : data index (n data points)
 j : k classes

$$C(\theta) = - \sum_{i=1}^m \sum_{j=1}^k \mathbb{I}\{y_i = j\} \log p(y=j | x_i, \theta)$$

" $e^{\theta^T x}$

(By Andrew Ng's UFLDL tutorial
webpage)

$$\frac{e^{\theta^T x}}{e^{\theta^{(1)T} x} + e^{\theta^{(2)T} x} + \dots + e^{\theta^{(k)T} x}}$$

Proofs you probably weren't taught

Friday, October 21, 2011

Why is the error function minimized in logistic regression convex?

We want to prove that the error/objective function of logistic regression :

$$J(\theta) = \sum_{i=1}^m y^i [-\log(h_{\theta}(x^i))] + (1 - y^i) [-\log(1 - h_{\theta}(x^i))] \quad (1)$$

$$\text{where } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

is convex.

Proof:

Before beginning the proof, i would first like to make you review/recollect a few definitions/rules/facts/results related to convex functions:

- **Definition of a convex function:** A function $f(x)$ is said to be convex if the following inequality holds true:

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \quad \forall x, y \in \text{Domain}(f) \text{ and } \alpha \in [0, 1]$$

- **First-order condition of convexity:** A function $f(x)$ which is differentiable is convex if the following inequality condition holds true:

$$f(y) \geq f(x) + \nabla_x^T f(x)(y - x); \quad \forall x, y \in \text{Domain}(f)$$

Intuitively, this condition says that the tangent/first-order-taylor-series approximation of $f(x)$ is globally an under-estimator.

- **Second-order condition of convexity:** A function $f(x)$ which is twice-differentiable is convex if and only if its hessian matrix (matrix of second-order partial derivatives) is positive semi-definite, i.e.

$$\forall z: z^T \nabla_x^2 f(x) z \geq 0 \text{ where } \nabla_x^2 f(x) \text{ is the hessian}$$

- **Sum/Linear-combination of two or more convex functions is also convex:** Let $f(x)$ and $g(x)$ be two convex functions. Then any linear combination of these two functions

$$(\lambda_1 f + \lambda_2 g)(x) = \lambda_1 f(x) + \lambda_2 g(x)$$

is also a convex function (this can be easily proved using the definition of the convex function).

Now notice that if we can prove that the two functions

$$-\log(h_{\theta}(x^i)) \text{ and } -\log(1 - h_{\theta}(x^i))$$

are convex, then our objective function

$$J(\theta) = \sum_{i=1}^m y^i [-\log(h_{\theta}(x^i))] + (1 - y^i) [-\log(1 - h_{\theta}(x^i))]$$

must also be convex since any linear combination of two or more convex functions is also convex.

Let us now try to prove that

About Me

 My photo

Deepak Roy Chittajallu

Curiosity is the mother of all invention !!!

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▼ [2011](#) (10)

▼ [October](#) (2)

[Proof for the first-order condition of convexity](#)

[Why is the error function minimized in logistic re...](#)

► [July](#) (3)

► [June](#) (5)

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$$-\log(h_\theta(x)) = -\log\left(\frac{1}{1+e^{-\theta^T x}}\right) = \log(1 + e^{-\theta^T x})$$

is a convex function of theta. In order to do this, we will use the second-order condition of convexity described above. Let us first compute the hessian matrix:

grad :

$$\begin{aligned}\nabla_\theta [-\log(h_\theta(x))] &= \nabla_\theta \left[\log(1 + e^{-\theta^T x}) \right] \\ &= \left(\frac{-e^{-\theta^T x}}{1+e^{-\theta^T x}} \right) x \\ &= \left(\frac{1}{1+e^{-\theta^T x}} - 1 \right) x \\ &= (h_\theta(x) - 1) x\end{aligned}$$

hessian :

$$\begin{aligned}\nabla_\theta^2 [-\log(h_\theta(x))] &= \nabla_\theta (\nabla_\theta [-\log(h_\theta(x))]) \\ &= \nabla_\theta ((h_\theta(x) - 1) x) \\ &= h_\theta(x) (1 - h_\theta(x)) x x^T\end{aligned}$$

Now below is the proof that this hessian matrix is positive semi-definite:

$$\begin{aligned}\forall z : z^T \nabla_x^2 (-\log(h_\theta(x))) &= z^T [h_\theta(x) (1 - h_\theta(x)) x x^T] z \\ &= h_\theta(x) (1 - h_\theta(x)) (x^T z)^2 \geq 0 \quad (2)\end{aligned}$$

Let us now try to prove that

$$\begin{aligned}-\log(1 - h_\theta(x)) &= -\log\left(1 - \frac{1}{1+e^{-\theta^T x}}\right) \\ &= -\log\left(\frac{e^{-\theta^T x}}{1+e^{-\theta^T x}}\right) \\ &= \theta^T x + \log(1 + e^{-\theta^T x})\end{aligned}$$

is a convex function of theta. In order to do this, we will again use the second-order condition of convexity described above. Let us first compute its hessian matrix:

grad :

$$\begin{aligned}\nabla_\theta [-\log(1 - h_\theta(x))] &= \nabla_\theta [\theta^T x + \log(1 + e^{-\theta^T x})] \\ &= x + \nabla_\theta [\log(1 + e^{-\theta^T x})]\end{aligned}$$

hessian :

$$\begin{aligned}\nabla_\theta^2 [-\log(1 - h_\theta(x))] &= \nabla_\theta (\nabla_\theta [-\log(1 - h_\theta(x))]) \\ &= \nabla_\theta \left(x + \nabla_\theta [\log(1 + e^{-\theta^T x})] \right) \\ &= \nabla_\theta^2 [-\log(h_\theta(x))] \\ &\quad \text{(we have proved in Eq. (2) above that this is positive semi-definite)}\end{aligned}$$

Above, we have proved that both

$$-\log(h_\theta(x^i)) \text{ and } -\log(1 - h_\theta(x^i))$$

are convex functions. And, the error/objective function of logistic regression

$$J(\theta) = \sum_{i=1}^m y^i [-\log(h_\theta(x^i))] + (1 - y^i) [-\log(1 - h_\theta(x^i))]$$

is essentially a linear-combination of several such convex functions. Now, since a linear combination of two or more convex functions is convex, we conclude that the objective function of logistic regression is convex.

Hence proved ...

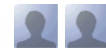
Following the same line of approach/argument it can be easily proven that the objective function of logistic regression is convex even if regularization is used.

Posted by Deepak Roy Chittajallu at 9:38 PM

Labels: Logistic Regression, Machine Learning

12 comments:

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